Facheng Yu

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EDUCATION

University of Washington, *Seattle*
Ph.D. in Statistics (Expected 2028)
M.S. in Statistics (2025)
Advisor: Zaid Harchaoui
Research Interests: Machine Learning, Causal Inference, Core AI, and AI Domain Applications
Coursework: Statistical Inference, Statistical Learning, Advanced Probability, Convex Optimization, Regression Methods

SKILLS

Programming: Python (NumPy, PyTorch, scikit-learn, Pandas), R, C++, SQL.
Tools: Git, LaTeX.

INDUSTRIAL EXPERIENCE

Microsoft Research Lab – New York City Jun 2026–Sep 2026
Research Intern, Machin Learning Group
Mentors: Pavel Metrikov, Miro Dudik

- Proposed, verified, and improved new offline evaluation methods for deterministic recommendation system using DCG.
- Tested a new objective defined for deterministic ranking policies which outperformed DCG with experimental evidence.

RESEARCH EXPERIENCE

Project: LLM-based Privacy-Preserving Authorship Obfuscation Sep 2025–Jun 2026

- Trained style classifiers on web resource data (e.g. Wikipedia) with well-constructed texts generated by LLMs (e.g. BERT).
- Finetuned LLMs (Qwen-32B) for obfuscation consisting of identifying, for a given author, its prominent stylistic features and perturbing the text along the corresponding invariant axes (aka, style axes) to bring their values closer to the average on the population.
- Evaluated the obfuscated text according to several metrics, such as linguistic acceptability (with CoLA models), coherence, fluency, quality and understandability.

Project: Multi-CLIP: Contrastive Learning for Multimodality Sep 2025–Jun 2026

- Generalized a multi-CLIP loss for n modalities as one step of balancing using known marginal distributions of each modality.
- Established a general learning chain for multimodality with theoretical guarantees inspired from iterative proportional fitting.
- Proposed an iterative algorithm that reduces the estimation variance of the multi-CLIP minimizer and performed experiments on three modalities (text, video, audio) in PyTorch.

Project: Stochastic Gradients under Nuisances Jun 2024–Dec 2025

- Investigated stochastic gradient algorithms for optimizing objectives with unknown nuisance parameters, with applications in high-dimensional inference and causal learning.
- Applied Neyman orthogonality to mitigates the bias introduced from nuisance estimation, enabling more reliable gradient-based optimization.
- Designed and implemented an approximately orthogonalized gradient oracle to debias learning for non-orthogonal objectives.
- Developed OSGD (Python library) to implement orthogonalized stochastic gradients for partially linear models (PLM), achieving improved risk consistency (~0.001) with ~0.1 nuisance error and ~0.5 operator error compared to standard SGD.

Project: Data Integration Using Covariate Summaries from External Sources May 2023–Jun 2024

- Designed robust mean outcome estimators under both homogeneous and heterogeneous data settings.
- Extended our integration methods to estimate treatment effects for improved **generalizability** and **transportability**.
- Conducted extensive simulations in R, demonstrating strong robustness to model misspecification and improved accuracy over baseline methods (e.g., DoubleML).

PUBLICATIONS

[1] **F. Yu**, R. Mehta, A. Luedtke, and Z. Harchaoui, "Stochastic Gradients under Nuisances", *NeurIPS 2025*.
[2] **F. Yu**, Z. Qi, and Y. Zhang, "Semi-Supervised Inference Using Unlabeled Summary Statistics", *arXiv:2411.15691*.